Your FREE School Teacher in Your Pocket! 📱 ✨

🎉 100% FREE • No Airtime • No Data • No Internet • Works Completely OFFLINE!

What is IsikoloAI? 🤔

JB³ ISIKOLO AI is like having a brilliant, patient teacher right on your phone, ready whenever you need help.

It can explain homework, guide your studying, and make tricky school work feel much easier.

Think of it as your friendly school buddy that never gets tired of helping you learn!

It's like having a magical homework whisperer, turning baffling assignments into a breeze!

Consider it your cheerful study pal 🤗

Why It's AMAZING! 🌟

Feature	Why You'll Love It
Works offline	No internet? No problem! Learn anytime, anywhere.
No cost ever	It stays completely free, so everyone can use it.
Speaks your language	Ask questions in the language that feels best for you.
Full CAPS syllabus	It covers the whole school journey from Grade R to Grade 12.
Patient teacher	It explains things calmly, again and again, until it clicks.
Always available	Any time you want to learn, JB^3 ISIKOLO AI is ready to help.

Everything You Need for School 📚

The entire South African CAPS curriculum is preloaded
from Grade R all the way to Grade 12!

That means every subject, every grade, and every lesson is ready to help you learn and grow. It's like carrying a whole classroom in your pocket, built just for you.

How It Works 🚀

1. Download the app on your phone.
2. Open it instantly no login needed.
3. Ask questions in your language.
4. Get help right away, anytime you need it.
5. Learn more and ace your tests with confidence!

No Barriers, Just Learning! 💪

❌ No Internet Needed
❌ No Data Costs
❌ No Airtime Required
❌ No Hidden Fees
✅ 100% FREE Forever

English, Afrikaans, isiZulu, isiXhosa, isiNdebele, Sepedi (Sesotho sa Leboa), Sesotho, Setswana, siSwati, Tshivenda, and Xitsonga

Your education, your language, your device , completely FREE! 🎓💝

iSiKoloAi

WHAT IS IT?

Category: On Device Edge AI / Educational Technology (EdTech)
Status: MultiPlatform Functional Prototype
Edition: Community Edition (Open Source) Supported
Platforms: iOS & Android Cross-Platform Deployment

JB^3 ISIKOLO AI is a state-of-the-art, multi-platform educational AI assistant designed specifically to operate under strict infrastructure constraints. By executing lightweight, highly optimized Large Language Models natively on standard commercial mobile hardware, ISIKOLO AI acts as a patient, encouraging, and brilliant local school teacher that fits right in a student's pocket.

The system comes preloaded with the full CAPS (Curriculum and Assessment Policy Statement) framework the official, centralized educational blueprint used by the South African Department of Basic Education (DBE) to guide teaching and learning from Grade R to Grade 12.

Crucially, the entire system is 100% offline, requiring zero data, zero internet access, and no remote cloud servers to deliver real-time, curriculum-aligned educational support. This platform is available completely FREE for anyone who has a compatible cell phone, computer, or device.

Your education, your language, your device , completely FREE! 🎓💝

WHY IT MATTERS

The digital divide in emerging markets is fundamentally driven by two barriers: volatile network connectivity and prohibitive cellular data costs. Traditional AI solutions rely entirely on expensive cloud infrastructures that systematically lock out the students who need interactive tutoring the most.

ISIKOLO AI shifts the paradigm by bringing advanced cognitive computing power directly to the device edge. By running entirely on-device, it guarantees:

- **Democratic Access:** High-quality educational support accessible anywhere—from remote rural communities to deep baseline classrooms without cell towers.
- **True Zero Ongoing Costs:** Once installed on a device, the application incurs absolutely no network data overhead for the student, school, or parent.
- **Absolute Privacy & Safety:** Conversations never leave the device, ensuring total child data safety, security, and protection by default.

KEY FEATURES

- **100% Air-Gapped Operation**: Core interaction pipelines, translation matrices, and inference layers operate entirely offline, shielded from connectivity drops.
- **Preloaded CAPS Knowledge Base:** Built-in repository of the official DBE curriculum, allowing the AI to ground its answers strictly within verified local academic standards from Grade R to Grade 12.
- **Multilingual Voice Interface Layer:** Integrated native, offline Speech-to-Text (STT) and text-to-speech (TTS) synthesis modules, enabling students to interact using natural voice inputs and receive spoken local responses.
- **The "Isikolo" Pedagogical Persona:** Custom-modeled to act as an empathetic, adaptive educator who translates complex science, technology, and mathematical jargon into simple, student-friendly analogies.
- **On-Device Vector RAG Guardrails:** Operates an embedded vector database to instantly index and pull exact CAPS framework syllabus snippets, keeping the AI response factually anchored to the official classroom syllabus.

TECHNICAL SPECIFICATIONS

ISIKOLO AI bridges standard web layout components with highly specialized, bare-metal hardware acceleration wrappers.

- **Core Framework Stack:** Engineered as a decoupled hybrid application using the Capacitor runtime bridge to seamlessly map lightweight frontend user interfaces directly to a compiled native C++ background worker.
- **The Native Inference Engine:**
  Powered by a low-level native compilation of the Llama.cpp framework embedded inside the mobile target runtime directories.
- **Google Gemma Core Logic (E2B - Effective 2B):** Features Google's state-of-the-art open-weights model optimizing 2.3 billion active parameters (5.1B total). Extensively quantized to compact its footprint into under 1.5GB to 3GB of RAM, allowing real-time execution on typical commercial smartphones.
- **Localized MzansiLM Module:** A highly efficient 125M parameter decoder-only language model optimized on local data corpora. Operating with a hidden size of 512, intermediate size of 1536, 30 layers, 9 attention heads, and a 2048 context length, it handles localized linguistic processing and translation tasks entirely on-device.
- **Hardware Layer Acceleration:** Fully wires into native platform performance matrices—engaging Apple Silicon's Metal execution layers on iOS and the Neural Networks API (NNAPI) on Android to maximize on-chip NPU and GPU hardware capabilities.

THE FOUR PILLARS OF THE ISIKOLO FRAMEWORK

The platform's user experience is anchored around four foundational pillars designed to maximize student engagement and community value:

- **Umhlolo Wesifundo (The Magic of Study):** Fostering direct, curiosity-driven interactive learning where abstract concepts are transformed into immediate, understandable offline insights.
- **Isixwayiso Sobalimi (The Guide for Growers / Mentorship):** Providing structured pedagogical guardrails and mental formatting that safely cultivate a student’s critical thinking capacity over time.
- **Ifundo Yomphakathi (Community Study):** Localizing educational context

Through multi-language compatibility (English, isiZulu, isiXhosa), ensuring learning honors shared social and cultural realities.

Imali Yesikolo (School Wealth/Capital): Preserving community resources and family capital by completely eliminating recurring data costs and reliance on hardware server fees.

USE CASES

- **Data-Free Homework Support:** Learners can deeply query physics equations, history timelines, or language structures at home late at night without requiring data tokens or parent cell contracts.
- **Connectivity-Independent Classrooms:** Teachers in schools with structural cellular reception dead zones can deploy the app as an active, localized teaching assistant to generate classroom quiz structures and cross-reference syllabus rules instantly.
- **Linguistic Scaffolding:** Students can formulate complex academic thoughts in their home languages, leveraging the offline dual-model pipeline to smoothly bridge comprehension gaps to English testing environments.

IsikoloAi is built. Now we are opening founding support to keep it free for South African learners and expand access.

It is a **free AI-powered education app that works offline without internet** created to support South African learners with schoolwork, study help, revision, explanations and practical learning support. SA CAPS syllabus Grades 5 - 12

The mission is simple: keep IsikoloAi free for learners in South Africa.

We are now opening founding support to help fund the next phase: AI credits, hosting, platform infrastructure, curriculum content, local language support, learner access and continued development.

International supporters, sponsors and partners can help make free access possible.

JB^3 ISIKOLO Ai

THE 10-STEP NEURAL PIPELINE: FROM CURIOSITY TO COMPREHENSION

THE NDEBELE CLASSROOM FLOW

1 10-STEP PIPELINE FLOW

1. THE SPARK – Learner Speaks
2. THE SCRIBE – Voice-to-Text
3. THE BRIDGE – Home Tongue to Eng
4. THE BLOCKS – Text Tokenization
5. THE TEXTBOOK – Local Vector DB
6. THE LESSON – Syllabus Fusion
7. THE BRAIN – Gemma Core Process
8. THE ANSWER – Simple Analogy
9. THE RETURN – Back to Home Tongue
10. VOICE – Offline TTS

2 TEN STEP CARDS

1 Learner Voice Input – The Spark of Curiosity

Technical: The smartphone microphone captures raw analog audio directly from the learner.

Classroom: A student asks a question aloud in isiZulu, isiXhosa, English, or another language.

Edge Advantage: Fully air-gapped. No voice recordings are sent to the cloud.

2 On-Device Speech-to-Text (STT) — Writing the Spoken Word

Technical: A lightweight acoustic model converts audio waves into text characters.

Classroom: *The phone acts like a fast classroom scribe, instantly writing down the spoken words.*

3 MzansiLM Translation Layer — The Language Bridge

Technical: MzansiLM translates local South African phrasing into English.

Classroom: *If the learner speaks in a home language, the local language helper translates it for the reasoning engine.*

4 Text Tokenization – Breaking into Learning Blocks

Technical: The English text is split into numeric tokens using a local vocabulary matrix.

Classroom: *The sentence is broken into small numbered puzzle blocks so the AI can read it piece by piece.*

5 Local Vector DB Query (RAG) — Consulting the CAPS Textbook

Technical: The tokens generate an embedding that queries an on-device database with official DBE guidelines.

Classroom: *Before answering, the AI checks its digital copy of the South African CAPS curriculum stored on the phone.*

6 Context Injection – Gathering the Lesson Plan

Technical: Relevant CAPS snippets are appended into the prompt memory stack.

Classroom: *The app stitches the learner's question together with the official lesson guidance into one guided lesson file.*

7 Google Gemma 2B Processing — The Teacher's Brain Sparks

Technical: *The consolidated prompt is processed by Google Gemma 2B running locally on the device.*

Classroom: *The AI teacher reads the question and CAPS rules together to work out the correct answer.*

Hardware Magic: On iPhone, this runs on-device via Apple Metal and the mobile GPU.

8 Educational Content Synthesis — Crafting the Perfect Analogy

Technical: *Gemma outputs a response stream guided by a strict pedagogical system prompt.*

Classroom: *The answer is written like a patient teacher: simple, visual, and easy to understand.*

9 MzansiLM Back-Translation — Returning to the Home Tongue

Technical: *The English answer is translated back into the learner's original language.*

Classroom: *The brilliant answer comes back in the student's home language so it feels natural and familiar.*

10 Offline Text-to-Speech (TTS) — The Classroom Voice Speaks

Technical: An on-device speech engine converts the final text into clear audio waveforms.

Classroom: *The phone speaks the answer aloud in a warm local voice. The lesson is complete, fully offline.*


JB3
100% FREE • No Airtime • No Data • No Internet • Works Completely OFFLINE
GRADE 3 TO 12
CAPS curriculum
SETTINGS
Server URL
capacitor://localhost
Engine
Gemma + MzansiLM (Translate & Think)
Language
English (SA)
Voice
Kate
Voice on
Diagnostic Terminal
(Enhanced), Noelle (Enhanced), Tessa, Daniel, Eddy, Flo, Grandma, Grandpa, Reed, Rocko, Sandy, Shelley, Karen, Samantha, Junior, Kathy, Whisper, Zarvox
[SYS] Onboarding: Jono, age 15, lang en
[SYS] Success! Initialized Context ID: 2 via path target: gemma-2-2b-it-q4_k_m.gguf
[SYS] Models loaded into RAM successfully!
Save & Reconnect
Reset Profile
19 voices for English (SA) · 19 on-device (· = best quality).
My neural engines are loaded into the A18 Pro chip. I am completely offline. How can I help you?
angavamile. Kusemqoka kakhulu u
abambe indawo yesibili kwi-log
yamaqoqo. I-Zambia yidlale kahle
kakhulu, yabamba kahle, yabamba kal
I-Zambia iyiqembu eliqinile. I-Zambia
yiqembu elinamandla kakhulu, eliqinile
kakhulu. Iqembu eliqinile kodwa elihle.
Gemma + MzansiLM (Translate & Think)
Language
isiZulu
Voice
English voic
Diagnostic T
Save &
No isiZulu T
speak respo
understand i
limitation.
English (SA)
Afrikaans
✓ isiZulu
isiXhosa
Sesotho
Sepedi
Xitsonga
Setswana
siSwati
Tshivenda
Sawubona! 👋
Let's set up your learning space.
Your name
e.g. Sipho
Age
e.g. 14
Home language
English (SA)
Continue
JB³ ISIKOLO AI
OFFLINE VOICE · ZA
Native Edge AI Ready
SETTINGS
i need help wirh math
Well, hello there! I'm ISIKOLO, and I'm here to help you with your math. Tell me, what kind of math is it that you're struggling with? Is it something specific, like fractions, decimals, or maybe algebra? Or is it just a general feeling of not understanding what's going on? The more details you give me, the better I can help! 😊
APPLE EDGE NATIVE (LLAMA.CPP FRAMEWORK)
Type a message...
Hold to talk
Download the App Now

Tshwaelo ya Lentšu la Moithuti : (Go theetša molaetša wa lentšu ntle le inthanete) 📱 ✨

100% MAHALA | Ga go nyakege Airtime | Ga go nyakege Data | Ga go nyakege Inthanete | E šoma ka ntle le inthanete (OFFLINE) ka botlalo!

IsikoloAi ke eng? 🤔

ISIKOLOAi e swana le go ba le morutiši yo a nago le tsebo le pelo e telele mogaleng wa gago, a ikemišeditše go thuša nako e nngwe le e nngwe ge o hloka thušo.

E ka hlaloša mešomo ya gae, ya hlahla go ithuta ga gago, gomme ya dira gore mešomo e thata ya sekolo e be bonolo kudu. Naganela gore ke mogwera wa gago wa sekolo wa go loka yo a sa lapišwego ke go go thuša go ithuta!

E swana le go ba le mothuši wa mehlolo wa mešomo ya gae, a fetola mešomo ye e gakanegago go ba ye bonolo kudu! E tšee bjalo ka mogwera wa gago wa go ithuta wa go thaba.

Ke Ka Lebaka La Eng E Le E MAKATSANG! 🌟

Karolo	Lebaka leo o tlago go e Rata
E šoma ntle le inthanete	Ga go inthanete? Ga go bothata! Ithute nako efe goba efe, lefelong lefe goba lefe.
Ga go na tefo le ka mohla	E dula e le mahala ka botlalo, gore mang le mang a kgone go e šomiša.
E bolela leleme la gago	Botšiša dipotšišo ka leleme leo le go loketšego gabotse.
Silabase e tletšego ya CAPS	E akaretša leeto la sekolo ka moka go tloga go Kereiti ya R go fihla go Kereiti ya 12.
Morutiši wa pelo e telele	E hlaloša dilo ka thedimošo, gape le gape, go fihlela o kwešiša.
E dula e le gona	Nako efe goba efe ge o nyaka go ithuta, JB³ ISIKOLO AI e ikemišeditše go thuša.

Se Sengwe le Se Sengwe Seo O Se Hlokago Bakeng Sa Sekolo📚

Kharikhulamo ya CAPS ya Afrika Borwa ka moka e tsentšwe go tloga go Kereiti ya R go fihla go Kereiti ya 12! Seo se ra gore thuto e nngwe le e nngwe, kereiti e nngwe le e nngwe,
gomme lešomo le lengwe le le lengwe di gona go go thuša go ithuta le go gola. E swana le go rwala phapoši ya borutelo ka moka ka pokothong ya gago, yeo e agetšwego wena.

Ka Mo E Šomago Ka Gona 🚀

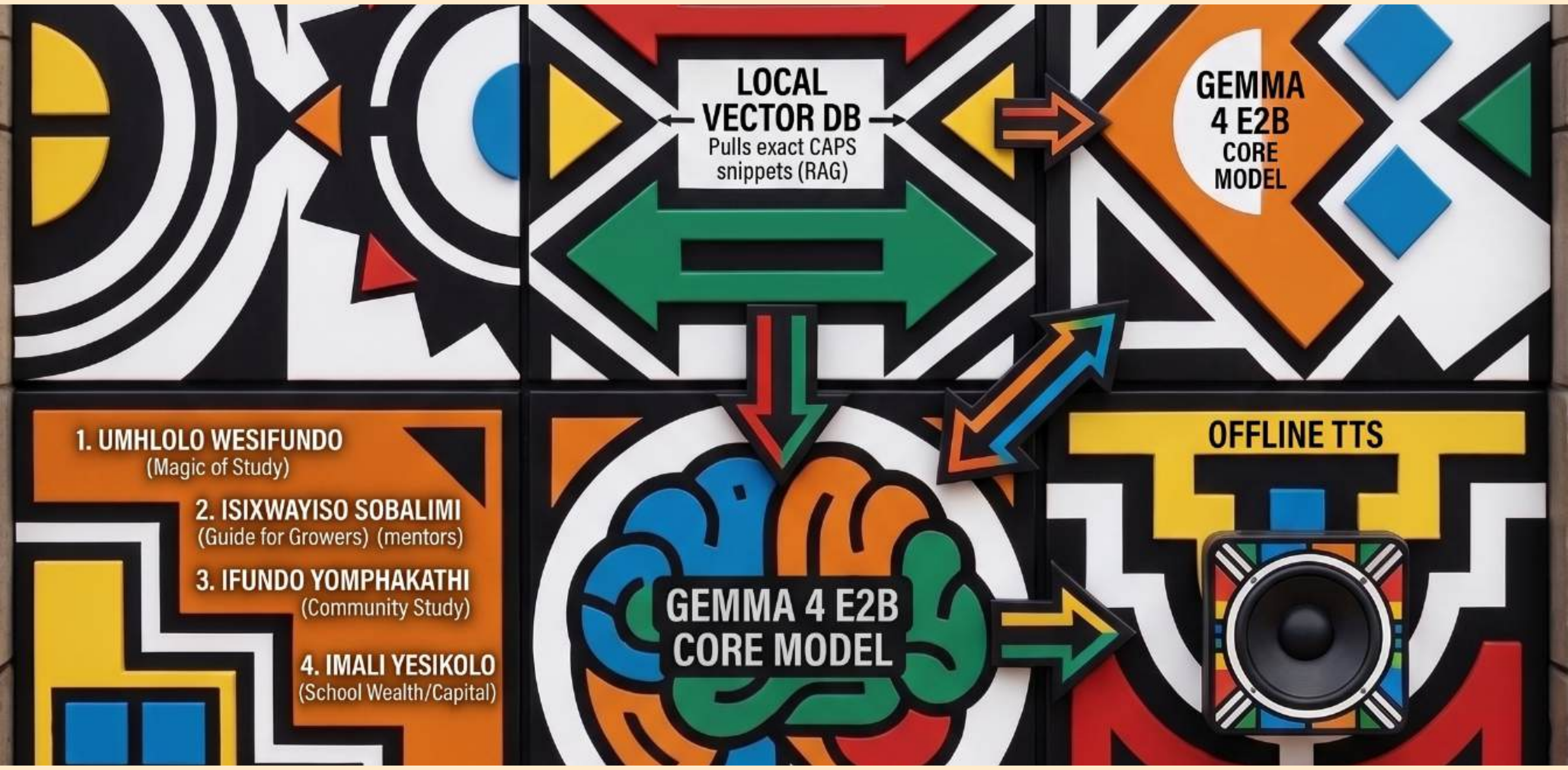
1. Taonelouta epe (app) mogaleng wa gago.
2. E bule gatee-tee - ga go nyakege gore o tsene (no login needed).
3. Botšiša dipotšišo ka leleme la gago.
4. Hwetša thušo gatee-tee, nako efe goba efe ge o e hloka.
5. Ithute kudu gomme o pase diteko tša gago ka boitshepo!

Ga go Mapheko, Ke Thuto Fela!💪

❌ Ga Go Nyakege Inthanete
❌Ga Go Ditshenyegelo tša Data
❌Ga Go Nyakege Airtime
❌Ga Go Ditefelo Tše Di Utamilego
✅100% MAHALA Go ya Go ile

English, Afrikaans, isiZulu, isiXhosa, isiNdebele, Sepedi (Sesotho sa Leboa), Sesotho, Setswana, siSwati, Tshivenda, and Xitsonga

Thuto ya gago, leleme la gago, sedirišwa sa gago, MAHALA ka botlalo!🎓💝



Se Sengwe le Se Sengwe Seo O Se Hlokago Bakeng Sa Sekolo

Kharikhulamo ya CAPS ya Afrika Borwa ka moka e tsentšwe go tloga go Kereiti ya R go fihla go Kereiti ya 12!

Seo se ra gore thuto e nngwe le e nngwe, kereiti e nngwe le e nngwe, gomme lešomo le lengwe le le lengwe di gona go go thuša go ithuta le go gola. E swana le go rwala phapoši ya borutelo ka moka ka pokothong ya gago, yeo e agetšwego wena.

🔧 Ka Mo E Šomago Ka Gona

1. **Taonelouta epe (app) mogaleng wa gago.**
2. **E bule gatee-tee - ga go nyakege gore o tsene (no login needed).**
3. **Botšiša dipotšišo ka leleme la gago.**
4. **Hwetša thušo gatee-tee, nako efe goba efe ge o e hloka.**
5. **Ithute kudu gomme o pase diteko tša gago ka boitshepo!**

💝 Ga go Mapheko, Ke Thuto Fela!

- Ga Go Nyakege Inthanete
- Ga Go Ditshenyegelo tša Data
- Ga Go Nyakege Airtime
- Ga Go Ditefelo Tše Di Utamilego
- 100% MAHALA Go ya Go ile

English, Afrikaans, isiZulu, isiXhosa, isiNdebele, Sepedi (Sesotho sa Leboa), Sesotho, Setswana, siSwati, Tshivenda, le Xitsonga

Thuto ya gago, leleme la gago, sedirišwa sa gago, MAHALA ka botlalo! 🎓💝

Dikokwane Tše Nne

- **UMHLOLO WESIFUNDO** (Mohlolo wa Thuto)
- **ISIXWAYISO SOBALIMI** (Tlhahlo ya Bahlahli / baeletši)
- **IFUNDO YOMPHAKATHI** (Thuto ya Setšhaba)
- **IMALI YESIKOLO** (Lehumo/Khamphatale ya Sekolo)

KE ENG?

- **Karolo:** AI ya mo Sedirišweng (On Device Edge AI) / Theknolotši ya Thuto (EdTech)
- **Boemo:** Setlwaedi sa go Šoma sa Dilaeborari tše Dintši (MultiPlatform Functional Prototype)
- **Tokollo: Tokollo ya Setšhaba** (Open Source) E Thekgilwe
- **Dilaeborari:** Phano ya Cross-Platform ya iOS le Android

DIKAROLO TŠE BOHLOKWA

- **100% Tshepedišo ye e Kgaotšwego go Inthanete (Air-Gapped):**
  Ditsela tša tshepetšo ya tirišano, phetolelo le go akanya di šoma ka moka ntle le inthanete, di sirilwe kgahlanong le go wa ga kgokagano.
- **Motheo wa Tsebo wa CAPS wo o Tsentšwego e sa le Pele:**
  Polokelo ye e agilwego ka gare ya kharikhulamo ya semmušo ya DBE, e dumelela AI go thea dikarabo tša yona godimo ga maemo a netefaditšwego a dithuto tša selegae go tloga go **Grade R go ya go Grade 12.**
- **Legae la Dipolelo Tše Dintši tša Lentšu:**
  Go kopantšwe dikarolo tša tlhago tša go fetolela segalontšu go ya go sengwalwa (STT) le sengwalwa go ya go segalontšu (TTS), go dumelela barutwana go dirišana ka lentšu la tlhago le go hwetša dikarabo tša segalontšu tša selegae.
- **Semelo sa Thuto sa "Isikolo":**
  Se dirilwe go šoma bjalo ka morutiši yo a nago le kwelobohloko, yo a kgonago go fetolela saense, theknolotši le mmetse ka mehlala ye bonolo ye e kwešišegago ke barutwana.
- **Ditšhireletšo tša go Boloka tša RAG tša mo Sedirišweng:**
  E diriša polokelo ya vector ye e tsentswego ka gare go tsenya ka pela le go hwetša dikarolwana tše nepagetšego tša kharikhulamo ya CAPS, e boloka dikarabo tša AI di theilwe godimo ga kharikhulamo ya semmušo ya ka klaseng.

TSHEPETŠO YA NEURAL YA DIKGATO TŠE 10: GO TLOGA GO FELA PELO GO YA GO KWEŠIŠO

TLHASE (The Spark) — Tshwaelo ya Lentšu la Moithuti

Moithuti o bolela potšišo goba tlhoko ka segalontšu goba sengwalwa. Tshepedišo e amogela tlhase ya mathomo ya go ithuta.

MONGWADI (The Scribe) — STT ya mo Sedirišweng

Segalontšu se fetolelwa go sengwalwa mo sedirišweng gore se be bonolo go se sekaseka. Se direga ntle le go romela data kgakala.

LEPOGO (The Bridge) — Legato la Phetolelo

Sengwalwa se beakanywa le go hlangwa ka leleme leo le swanetšego, gore tshepedišo e kwešiše potšišo ka mokgwa wo o nepagetšego.

DIPLOKO (The Blocks) — Go arola sengwalwa

Tše dingwe tša mantšu le dikgopolo di aroganywa ka diboloko tša tlhaeletsano, e le gore mohlala o kgone go šoma ka botlalo.

PUKU YA THUTO (The Textbook) — Databeisi ya Selegae

Tsebo ya CAPS le ditšupiso tša thuto di hwetšwa go databeisi ya selegae, di lokišeletšwe go araba ka nepagalo.

THUTO (The Lesson) — Tsenyo ya Taba

Potšišo ya moithuti e tsenywa ka gare ga tshepedišo ya tlhatlhobo gore e laole moo go nyakegago karabo.

BJOKO (The Brain) — Tshepedišo ya Google Gemma 2B

Mohlala wa pušo ya tlhago o sekaseka tshedimošo gomme o age karabo ka tsela ye e loketšego boemo bja moithuti.

KARABO (The Answer) — Go Hlama Karabo

Karabo e ahlwa ka mantšu ao a hlwekilego, ao a utollago kgopolo ka polelo ye bonolo le ya tirišo.

GO BOA (The Return) — Go Boela go Leleme la Gae

Ge go nyakega, karabo e bušetšwa morago go leleme la gabo moithuti bakeng sa kwešišo e botse.

LENTŠU (Voice) — Sengwalwa-go-Polelo

Karabo ya bofelo e fetolelwa gape go segalontšu gore moithuti a e kwe ka tlhago le ka bonolo.

KE KA LEBAKA LA ENG E LE BOHLOKWA

Lekgapo la dijithale mebarakeng ye e golago le hlolwa kudu ke mapheko a mabedi: kgokagano ye e sa tsepamago ya neteweke le ditshenyegelo tše di phagamego tša data ya megala. Ditharollo tša setšo tša AI di ithekgile ka botlalo go mananeokgoparara a leru a theko e phagamego ao a tswalelelago ka ntle baithuti bao ba hlokago kudu tlhatlhlo ya thuto ye e nago le tirišano.

ISIKOLOAi e fetoša mokgwa wo ka go tliša maatla a tseneletšego a khomphutha ya go nagana thwii moraleng wa sedirišwa. Ka go šoma ka botlalo mo sedirišweng, e netefatša:

- **Phihlelelo ya Temokrasi:**
  Thekgo ya thuto ya maemo a godimo yeo e fihlelelegago kae le kae - go tloga metseng ya magaeng ya kgole go ya go diphaphoši tša borutelo tša motheo ka ntle le ditora tša disele.
- **Ga go na Ditshenyegelo tša Nnete tše di Tšwelago pele:**
  Ge e se e tsentšwe (installed) mo sedirišweng, tirišo ga e be le tšenyo ya data ya neteweke go moithuti, sekolo, goba motswadi.
- **Bosephiri le Polokego ye e Tletšego:**
  Dipoledišano ga di tsoge di tlogile sedirišweng, go netefatša polokego ya data ya ngwana ka botlalo, tšhireletšo, le tšhireletšo ka thago.

DIKAROLO TŠE BOHLOKWA

- **100% Tshepedišo ye e Kgaotšwego go Inthanete (Air-Gapped):** Ditsela tše kgolo tša tirišano, dimaterikisi tša phetolelo, le magato a tshedimošo di šoma ka botlalo ka ntle le inthanete, di šireleditšwe go go kgaoga ga kgokagano.
- **Motheo wa Tsebo wa CAPS wo o Tsentšwego e sa Pele:** Bobolokelo bjo bo agetšwego ka gare bja kharikhulumo ya semmušo ya DBE, bjo bo dumelelago AI go thea dikarabo tša yona ka thata ka gare ga maemo a semmušo a thuto a selegae go tloga go Kereiti ya R go fihla go Kereiti ya 12.
- **Legae la Dipolelo Tše Dintši tša Lentšu:** Mmojule wa tlhago wo o kopantšwego, wa ka ntle le inthanete wa Polelo-go-Sengwalwa (STT) le thulaganyo ya sengwalwa-go-polelo (TTS), go dumelela baithuti go dirišana ka go šomiša mantšu a tlhago le go amogela dikarabo tša selegae tše di bolelwago.
- **Semelo sa Thuto sa "Isikolo":** Se hlamilwe go šoma bjalo ka morutiši wa kwelobohloko, wa go feto-fetoga le maemo yo a fetolelago mareo a raraganego a saense, theknolotši, le mmetse go ba ditshwantšhišo tše bonolo, tša go ba botšwalle go baithuti.
- **Ditšhireletšo tša go Boloka tša RAG tša mo Sedirišweng:** E šomiša databeisi ye e agetšwego ka gare go hwetša le go goga gatee-tee dikarolwana tša silabase tša tlhako ya CAPS, go boloka karabo ya AI e theilwe go nnete go silabase ya semmušo ya phapoši ya borutelo.

TSHEPETŠO YA NEURAL YA DIKGATO TŠE 10: GO TLOGA GO FELA PELO GO YA GO KWEŠIŠO

TLHASE (The Spark): Tshwaelo ya Lentšu la Moithuti
Moithuti o botšiša potšišo ka go hlaboša lentšu ka Sezulu, Sethosa, Seisemane, goba leleme le lengwe. E šoma ka ntle le inthanete ka botlalo; ga go dikgatiso tša lentšu tšeo di romelwago go leru.

MONGWADI (The Scribe): STT ya mo Sedirišweng (Lentšu-go-Sengwalwa)
Mogala o šoma bjalo ka mongwadi wa phapoši ya borutelo wa lebelo, o ngwala mantšu ao a boletšwego gatee-tee.

LEPOGO (The Bridge): Legato la Phetolelo la MzansiLM
Ge e le gore moithuti o bolela ka leleme la gae, mothuši wa leleme la selegae o

fetolela se go enjene ya go nagana (English).

DIPLOKO (The Blocks): Go arola sengwalwa (Text Tokenization)
Lefoko le arolwa go ba diploko tše nnyane tša go rarabolla tše di nago le dinomoro e le gore AI e kgone go e bala karolo ka karolo.

PUKU YA THUTO (The Textbook): Databeisi ya Selegae (Local Vector DB Query)
Pele e araba, AI e lekola khopi ya yona ya dijithale ya kharikhulamo ya CAPS ya Afrika Borwa yeo e bolokilwego mogaleng.

THUTO (The Lesson): Tsenyo ya Taba (Context Injection)
App e ruma potšišo ya moithuti gotee le tlhahlo ya thuto ya semmušo go ba faele e tee ya thuto ye e hlahliwago.

BJOKO (The Brain): Tshepedišo ya Google Gemma 2B
Morutiši wa AI o bala potšišo le melao ya CAPS gotee go rarabolla karabo ya maleba, e šoma ka theknolotši ya mogala ka ntle le inthanete.

KARABO (The Answer): Go Hlama Karabo ye Bonolo
Karabo e ngwalwa bjalo ka morutiši wa pelo e telele: e bonolo, ya ponagalo, gomme e kwagala gabonolo.

GO BOA (The Return): Go Boela go Leleme la Gae (MzansiLM Back-Translation)
Karabo ye e kgahlišago e boa ka leleme la gae la moithuti gore e kweagale e le ya tlhago gomme e tlwaelegile.

LENTŠU (Voice): Sengwalwa-go-Polelo (TTS) sa ka ntle le inthanete
Mogala o bolela karabo ka go hlaboša ka lentšu le borutho la selegae. Thuto e phethilwe, ka ntle le inthanete ka botlalo.

JB^3 ISIKOLO Ai

DIKAKANYO TŠA THEKNOLOTŠI

ISIKOLOAi e kopanya dikarolo tša setlwaelo tša peakanyo ya wepe le di-wrapper tša kgašo ya hadewere ye e ikgethilego kudu, ye e šomago ka thwii ka sedirišweng.

- **Core Framework Stack:**
  E hlamilwe bjalo ka tirišo ya lebasetere ye e arogantšwego ye e šomago ka Capacitor runtime bridge go mapša gabotse dikgokagano tša mosebediši wa ka pele ka go toba go modirišani wa ka morago wa C++ wa tlhago.
- **The Native Inference Engine:**
  E matlafaditšwe ke tlhakanyo ya tlhago ya maemo a tlase ya foreimi ya Llama.cpp ye e beilwego ka gare ga tirelo ya difaele tša runtime ya sedirišwa sa mogala.
- **Google Gemma Core Logic (E2B - Effective 2B):**
  E na le mohlala wa maemo a godimo wa open-weights wa Google o šomago ka 2.3 bilione ya di-parameter tše di šomago (5.1B ka moka). E quantized ka botlalo go fokotša sebaka sa yona go ka fase ga 1.5GB go ya go 3GB ya RAM, e dumelela tirišo ya nako ya nnete go di-smartphone tše tlwaelegilego tša kgwebo.
- **Localized MzansiLM Module:**
  Mohlala wa polelo wa decoder-only wa 125M parameter wo o diretšwego go šoma gabotse godimo ga dikgoboketšo tša data tša selegae. Wo šoma ka hidden size ya 512, intermediate size ya 1536, mekgahlelo ya 30, dihlogo tša tlhokomelo tše 9, le botelele bja context bja 2048, gomme o swaragana le tshepedišo ya polelo ya selegae le mediro ya phetolelo ka moka ka go sedirišwa.
- **Hardware Layer Acceleration:**
  E kopantšwe ka botlalo le dimetara tša tshepedišo tša sethala sa tlhago—e diriša mekhahlelo ya Metal execution ya Apple Silicon go iOS le Neural Networks API (NNAPI) go Android go oketša bokgoni bja NPU le GPU tša ka gare.

DIKOKWANE TŠE NNE TSA FOREIMI YA ISIKOLO

Maitemogelo a modiriši wa sefala a theilwe go dikokwane tše nne tša motheo tše di hlametšwego go oketša go kgatha tema ga moithuti le boleng bja setšhaba:

- **Umhlolo Wesifundo (The Magic of Study):**
  Go godiša thuto ya tshwaragano ye e tutuetšwago ke kganyogo, moo dikgopolo tše di sa bonagalego di fetošwago go ba ditemošo tša ka pela tše di kwešišegago, tša go šoma le ge go se na inthanete.
- **Isixwayiso Sobalimi (The Guide for Growers / Mentorship):**
  Go fa ditšhupetšo tša thuto tše di hlophilwego gabotse le tlhaka ya monagano ye e godišago bokgoni bja moithuti bja go nagana ka tshepedišo ya nako e telele ka tsela ye e šireletšegilego.
- **Ifundo Yomphakathi (Community Study):**
  Go loka ditaba tša thuto go ya le maemo a selegae le go dira gore dithuto di sepelelane le ditlhoko tša tikologo le ditshwantšho tša setšo go matlafatša kwešišo.
- **Imali Yesikolo (Lehumo/Khamphatale ya Sekolo):**
  Go fa baithuti bokgoni bja go ithekga ka thuto, go fokotša go ithekga ka diseva tša leru tše di jago tšhelete, le go netefatša gore thuto ya maemo a godimo e dula e le ya mahala e bile e fihlelega.

DIKOKWANE TŠE NNE TSA FOREIMI YA ISIKOLOq

Maitemogelo a modiriši wa sefala a theilwe go dikokwane tše nne tša motheo tše di hlametšwego go oketša go kgatha tema ga moithuti le boleng bja setšhaba:

- **Umhlolo Wesifundo (Mohlolo wa Thuto):** Go godiša thuto ya tlhamalalo, ye e tšweletšwago ke kganyogo moo dikgopolo tše di sa bonagalego di fetošwago go ba dikgakanego tše di kwešišegago ka pela, tša ka ntle le inthanete.
- **Isixwayiso Sobalimi (Tlhahlo ya Bahlahli / Baeletši):** Go fa ditaelo tša thuto tše di hlametšwego le go hlama ka monagano tšeo di hlahliša bokgoni bja go nagana bja moithuti ka tsela ye e bolokegilego ka nako.
- **Ifundo Yomphakathi (Thuto ya Setšhaba):** Go dira gore ditaba tša thuto di sepelelane le selegae le go kopanya tsebo ya setšo le ditshwantšhišo go godiša kwešišo ye e tseneletšego.
- **Imali Yesikolo (Lehumo/Khamphatale ya Sekolo):** Go fa baithuti le dikolo bokgoni bja go ikemela ka thuto, go fokotša go ithekga ka diseva tša leru tše di bitšago tšhelete le go netefatša gore thuto ya maemo a godimo e dula e le mahala le go fihlelega.

Go theetša molaetša wa lentšu ntle le inthanete Go theetša molaetša wa lentšu ntle le inthanete): STT ya mo Sedirišweng: E fetolela polelo go ba sengwalwa

Imali Yesikolo (School Wealth/Capital): Preserving community resources and family capital by completely eliminating recurring data costs and reliance on hardware server fees.

USE CASES

- **Thekgo ya Mešomo ya Gae e se nago Data:**
  Baithuti ba ka botšiša ka botebo dipotšišo tša fisiks, methalo ya histori, goba dibopego tša polelo gae bošego bja marega ntle le go nyaka ditokene tša data goba dikonteraka tša disele tša batswadi.
- **Diphaposi tša Sekolo tše di Ikemetšego go Kgokagano:**
  Barutiši ba dikolong tše di nago le mafelo a go hwa ga kgokagano ya disele ba ka tsenya tirišong app bjalo ka mothuši wa go ruta wa selegae, wa go šoma, go hlama dibopego tša diteko tša phaposi le go lekola melao ya silabase ka pela.
- **Thekgo ya Polelo:**
  Baithuti ba ka bopa dikgopolo tše di raraganego tša thuto ka dipolelo tša bona tša gae, ba šomiša mokgwa wa dimotšhene tše pedi wa ka ntle le inthanete go kopanya ka bonolo dikgoba tša kwešišo go ya go mafelo a diteko a Seisemane.

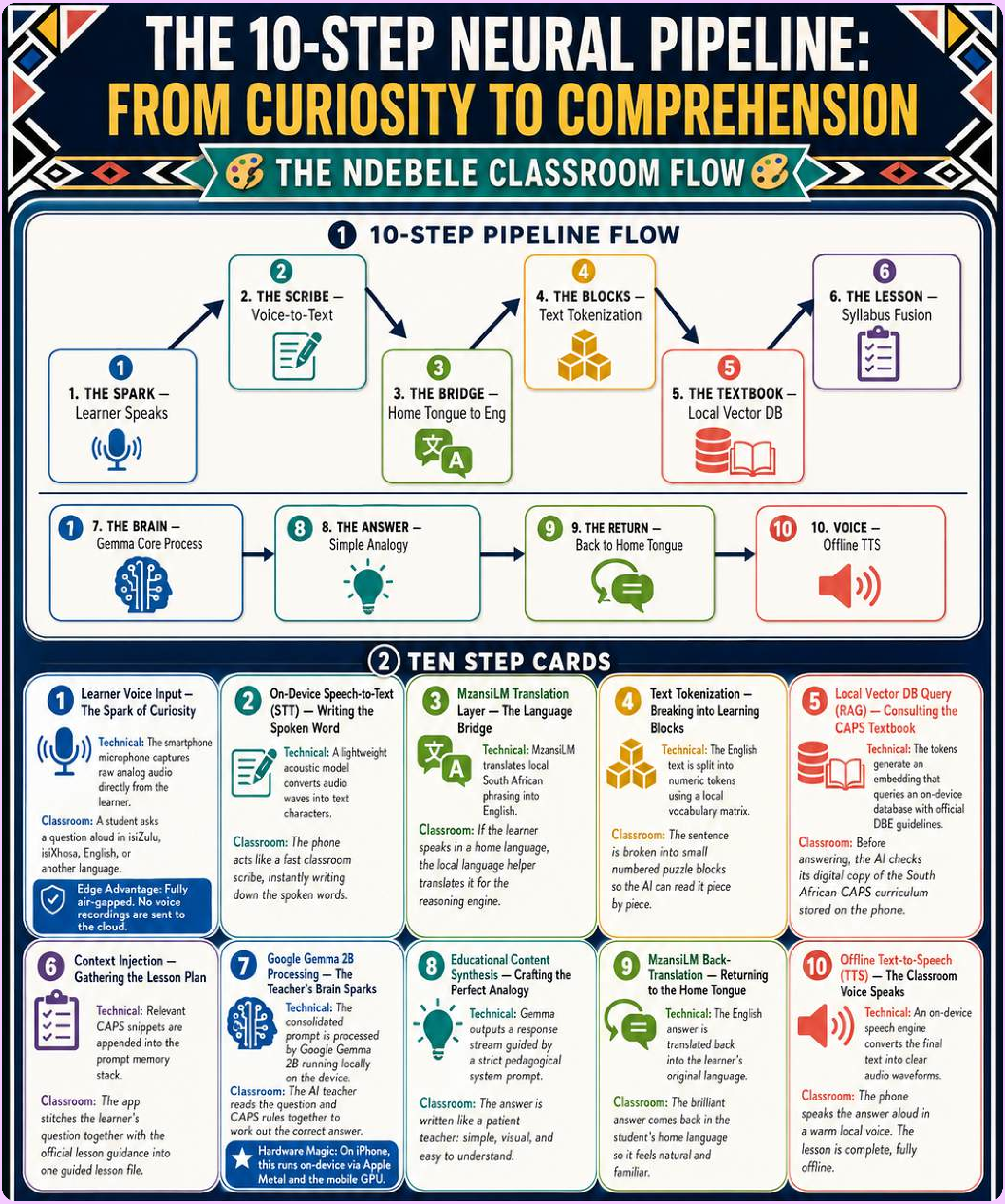
IsikoloAi e agilwe. Bjale re bula thekgo ya motheo go boloka e le mahala go baithuti ba Afrika Borwa le go katološa phihlelelo.

Ke app ya thuto ya AI ya mahala yeo e šomago ka ntle le inthanete yeo e hlametšwego go thekga baithuti ba Afrika Borwa ka mešomo ya sekolo, thušo ya go ithuta, boeletšo, ditlhalošo le thekgo ya go ithuta ya maleba. Silabase ya CAPS ya SA Dikereiti tša 5 - 12

Morero o bonolo: boloka IsikoloAi e le mahala go baithuti ba Afrika Borwa.

Bjale re bula thekgo ya motheo go thuša go lefela kgato ye e latelago: dikheorediti tša AI, go tshwara, motheo wa sefala, diteng tša kharikhulamo, thekgo ya polelo ya selegae, phihlelelo ya baithuti le tlhabollo ye e tšwelago pele.

Bathekgi ba boditšhabatšhaba, bafani le balekane ba ka thuša go dira gore phihlelelo ya mahala e kgonege.


THE 10-STEP NEURAL PIPELINE:
FROM CURIOSITY TO COMPREHENSION
THE NDEBELE CLASSROOM FLOW
1 10-STEP PIPELINE FLOW
1. THE SPARK – Learner Speaks
2. THE SCRIBE – Voice-to-Text
3. THE BRIDGE – Home Tongue to Eng
4. THE BLOCKS – Text Tokenization
5. THE TEXTBOOK – Local Vector DB
6. THE LESSON – Syllabus Fusion
7. THE BRAIN – Gemma Core Process
8. THE ANSWER – Simple Analogy
9. THE RETURN – Back to Home Tongue
10. VOICE – Offline TTS
2 TEN STEP CARDS
1 Learner Voice Input – The Spark of Curiosity
Technical: The smartphone microphone captures raw analog audio directly from the learner.
Classroom: A student asks a question aloud in isiZulu, isiXhosa, English, or another language.
Edge Advantage: Fully air-gapped. No voice recordings are sent to the cloud.
2 On-Device Speech-to-Text (STT) – Writing the Spoken Word
Technical: A lightweight acoustic model converts audio waves into text characters.
Classroom: The phone acts like a fast classroom scribe, instantly writing down the spoken words.
3 MzansiLM Translation Layer – The Language Bridge
Technical: MzansiLM translates local South African phrasing into English.
Classroom: If the learner speaks in a home language, the local language helper translates it for the reasoning engine.
4 Text Tokenization – Breaking into Learning Blocks
Technical: The English text is split into numeric tokens using a local vocabulary matrix.
Classroom: The sentence is broken into small numbered puzzle blocks so the AI can read it piece by piece.
5 Local Vector DB Query (RAG) – Consulting the CAPS Textbook
Technical: The tokens generate an embedding that queries an on-device database with official DBE guidelines.
Classroom: Before answering, the AI checks its digital copy of the South African CAPS curriculum stored on the phone.
6 Context Injection – Gathering the Lesson Plan
Technical: Relevant CAPS snippets are appended into the prompt memory stack.
Classroom: The app stitches the learner's question together with the official lesson guidance into one guided lesson file.
7 Google Gemma 2B Processing – The Teacher's Brain Sparks
Technical: The consolidated prompt is processed by Google Gemma 2B running locally on the device.
Classroom: The AI teacher reads the question and CAPS rules together to work out the correct answer.
Hardware Magic: On iPhone, this runs on-device via Apple Metal and the mobile GPU.
8 Educational Content Synthesis – Crafting the Perfect Analogy
Technical: Gemma outputs a response stream guided by a strict pedagogical system prompt.
Classroom: The answer is written like a patient teacher: simple, visual, and easy to understand.
9 MzansiLM Back-Translation – Returning to the Home Tongue
Technical: The English answer is translated back into the learner's original language.
Classroom: The brilliant answer comes back in the student's home language so it feels natural and familiar.
10 Offline Text-to-Speech (TTS) – The Classroom Voice Speaks
Technical: An on-device speech engine converts the final text into clear audio waveforms.
Classroom: The phone speaks the answer aloud in a warm local voice. The lesson is complete, fully offline.

From Prototype to Public Impact

The JB^{3} product ecosystem has moved beyond concept.
OS^{3}, Clipboard Ai, InvestigatorAi, VoiceGrid Ai, IsikoloAi, Newsroom, and DukeBox of London represent a working catalogue of practical AI-powered systems built for education, communication, intelligence, media, business operations, voice, music, and secure collaboration.

The next phase is deployment.
Partner, Sponsor, or Build With Us
JB^{3} is now opening conversations with clients, sponsors, technology partners, education partners, media partners, and organisations that want to support practical AI systems with real-world value.
Work With JB^{3}

Book a consultation for business diagnostics, operational systems, dashboards, automation, AI-supported workflows, and custom product development.
Sponsor the Final Launch Stage launch
we need AI credits, hosting, sound engineering, testing, deployment, annual software tools, learner access, or infrastructure.

Request a Product Demo
View the JB^{3} product ecosystem in action and explore which systems align with your business, school, publication, community,
or organisation.

Follow the Build
Subscribe for product updates, launch notes, YouTube videos, newsletters, case studies, and behind-the-scenes development.
Contact Jonathan Blackburn Founder JB^{3}

Website : www.jb3ai.com or the founder www.jonoblackburn.com
Email: hi@jb3ai.com or jono@jonoblackburn.com
WhatsApp: +277196918498
YouTube:_ @jb3ai @JonoBlackburn
Newsletter: hi@jb3ai.com

The systems are built. Now the mission is to turn working prototypes into public impact.

From Prototype to Public Impact

The JB³ product ecosystem is built. The next phase is deployment, sponsorship, testing, and scale.

Work With JB³

Business diagnostics, dashboards, automation, AI workflows, and custom product development.

Book a Consultation

Sponsor the Final 5%

Support launch needs such as AI credits, hosting, sound engineering, testing, annual software tools, learner access, and infrastructure.

Sponsor a Project

Request a Product Demo

View OS³, Clipboard Ai, InvestigatorAi, VoiceGrid Ai, IsikoloAi, Newsroom, and DukeBox of London in action.

Request a Demo

Follow the Build

Subscribe for product updates, launch notes, YouTube videos, newsletters, case studies, and behind-the-scenes development.

Subscribe for Updates

Jonathan Blackburn
Founder, JB³
Website: [ADD WEBSITE]
Email: [ADD EMAIL]
WhatsApp: [ADD NUMBER]
YouTube: [ADD LINK]
Newsletter: [ADD LINK]
Sponsor / Support Page: [ADD LINK]

Website

Sponsor Page

Book a Demo